

Auto Ticket Classification using Flow Designer

Platform: ServiceNow (Flow Designer) | **Domain:** School IT Helpdesk | **Approach:** No-code automation

1. INTRODUCTION

Project Title

Auto Ticket Classification using Flow Designer

Team Members

S.No	Name	Role
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1.1 Project Overview

The school IT helpdesk receives many incident requests every day from students and teachers: Wi-Fi problems, projector failures, forgotten passwords and slow computers. This project uses **ServiceNow Flow Designer** to classify each incident automatically at the moment it is created. A flow reads the **Short description** and **Description** fields, looks for keywords, and sets the **Category** (and optionally the assignment group) without any human action.

1.2 Purpose

- Classify incidents automatically at creation time.
- Reduce the manual triage effort of IT agents.
- Improve ticket routing speed and accuracy.
- Deliver a no-code solution that non-developers can maintain.

2. IDEATION PHASE

2.1 Problem Statement

IT staff currently read every incoming request and assign a category by hand. This is slow, inconsistent between agents, and delays routing to the right support team, especially during busy periods such as the start of term or exam weeks.

Who	School IT helpdesk agents; students and teachers raising tickets
Problem	Manual categorisation is time-consuming and inefficient
Impact	Delayed routing, inconsistent categories, higher resolution time

Desired outcome Tickets are categorised automatically when created

2.2 Empathy Map Canvas

Quadrant	IT Agent	Student / Teacher
Says	"I spend half my time just sorting tickets."	"I just need my projector working for class."
Thinks	"Many tickets are repetitive and easy to classify."	"Will anyone even look at my request soon?"
Does	Reads each ticket, picks a category, assigns a group	Submits a ticket and waits
Feels	Overloaded, frustrated by repetitive work	Anxious about delays, uncertain about status
Pain points	Manual triage, inconsistent categories	Slow response, unclear routing
Gains	Less admin work, more time for fixing issues	Faster resolution, confidence in the process

2.3 Brainstorming

Ideas considered:

- 1 Ask users to pick a category on the form (users often choose wrongly).
- 2 Business Rule with server-side script (works, but needs coding and is harder to maintain).
- 3 **Flow Designer with keyword conditions (selected)**: no-code, visible, easy to edit.
- 4 Machine-learning based Predictive Intelligence (powerful, but heavy for a school's volume).

Selected idea: a Flow Designer flow triggered on incident creation that uses keyword "contains" conditions to set the category.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User action	System behaviour	Outcome
Awareness	Faces an IT issue (e.g. Wi-Fi down)	n/a	Decides to raise a ticket
Submission	Creates incident with short description	Incident record created	Ticket logged
Classification	n/a	Flow triggers and classifies by keywords	Category/group set automatically
Assignment	n/a	Ticket routed to the right team	Agent sees a pre-categorised ticket
Resolution	Waits for update	Agent resolves	Issue fixed, ticket closed

3.2 Solution Requirement

Functional requirements

- FR1: Trigger automatically when an incident is created.
- FR2: Read keywords from Short description and Description.
- FR3: Set Category according to matched keywords.
- FR4: Apply a default category when no keyword matches.
- FR5: (Optional) Set the Assignment group for the matched category.

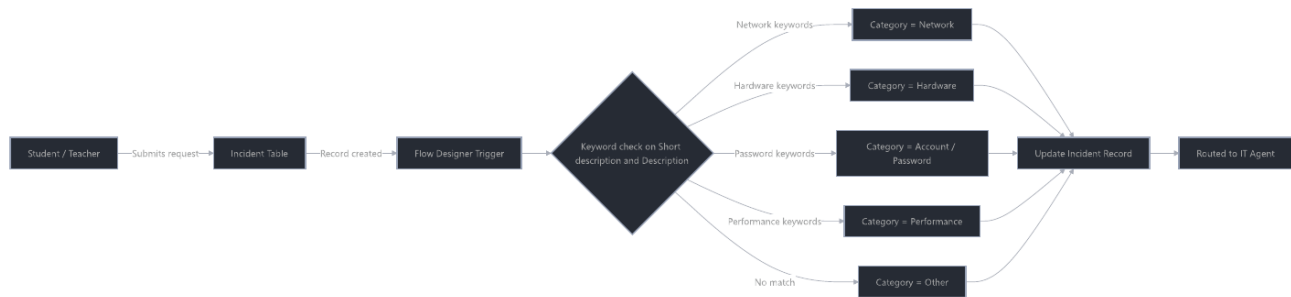
Non-functional requirements

- NFR1: No scripting; built entirely with Flow Designer actions.
- NFR2: Easy to add or edit keywords.
- NFR3: Runs without noticeable delay on ticket creation.

Keyword to category mapping

Category	Keywords (case-insensitive)	Suggested group
Network	wifi, wi-fi, internet, network, router, lan	Network Support
Hardware	projector, printer, monitor, keyboard, mouse, display	Hardware Support
Account / Password	password, login, locked, reset, account, access	Service Desk
Performance	slow, lag, hang, freeze, not responding	Desktop Support
Other (default)	no match	Service Desk

3.3 Data Flow Diagram



3.4 Technology Stack

Component	Technology
Platform	ServiceNow (Personal Developer Instance)
Automation	Flow Designer
Data	Incident table (incident)
Trigger	Created (Record trigger)
Logic	Flow Logic: If / Else If / Else
Action	Update Record

4. PROJECT DESIGN

4.1 Problem Solution Fit

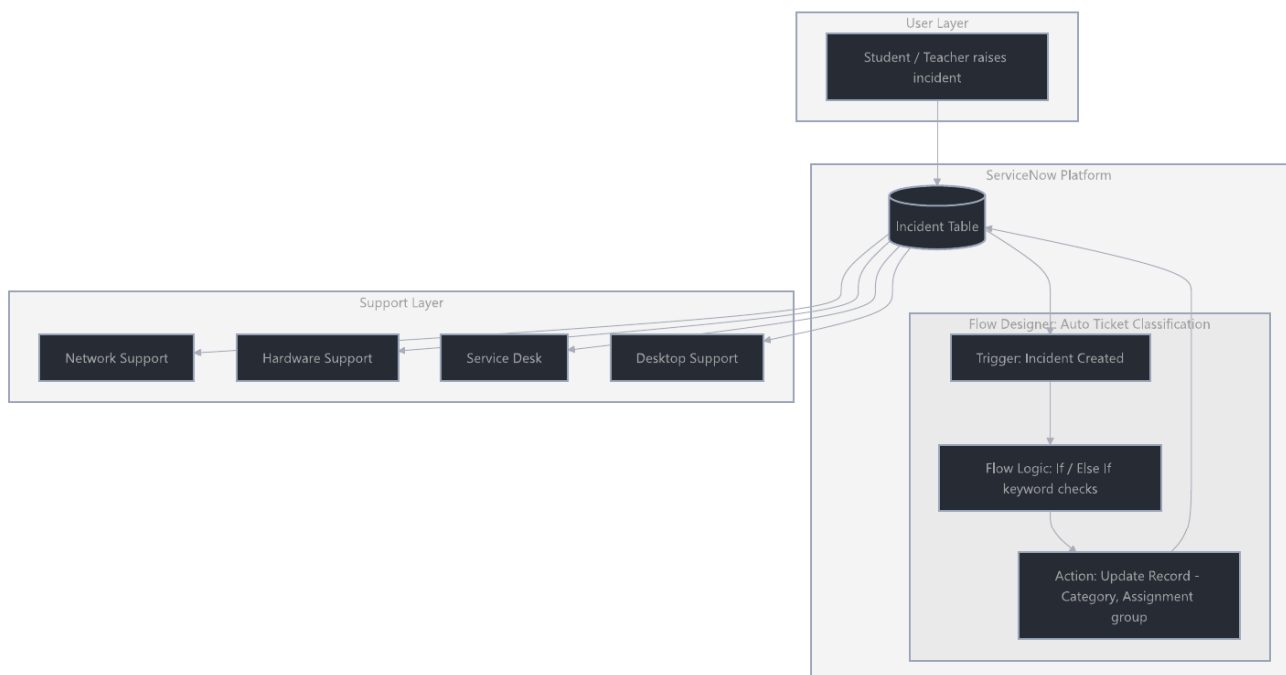
Problem	Solution	Fit
Manual categorisation takes time	Flow classifies at creation	Removes manual step
Inconsistent categories	Fixed keyword rules	Same input gives same category
Slow routing	Optional group assignment in the flow	Faster hand-off to correct team
Hard to maintain code	No-code Flow Designer	Admins can edit conditions visually

4.2 Proposed Solution

Create a flow named **Auto Ticket Classification** that:

- 1 Triggers when a record is **Created** on the **Incident** table.
- 2 Uses **If / Else If** logic to check whether *Short description* **contains** or *Description* **contains** a keyword group (OR condition).
- 3 Runs an **Update Record** action on the trigger record to set the Category (and optionally the Assignment group).
- 4 Falls through to an **Else** branch that sets the default category.
- 5 Is **Activated** so it runs on every new incident.

4.3 Solution Architecture



Build steps

- 1 Navigate to **All > Process Automation > Flow Designer** and click **New > Flow**. Name it *Auto Ticket Classification*.
- 2 Add trigger **Created** with table **Incident**.
- 3 Add **Flow Logic > If**. Condition: Short description *contains* wifi OR Description *contains* wifi (repeat for each Network keyword).
- 4 Inside the If, add action **Update Record**: Record = *Trigger > Incident Record*; Category = *Network*.
- 5 Add **Else If** branches for Hardware, Account / Password and Performance.
- 6 Add a final **Else** that sets Category = *Other* (default).
- 7 Click **Save**, then **Activate**.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activity	Duration
1	Requirement gathering and problem definition	Day 1
2	Keyword and category mapping	Day 2
3	Flow design (trigger and conditions)	Day 3 to 4
4	Update Record actions and default branch	Day 5
5	Testing with sample incidents	Day 6
6	Documentation and review	Day 7

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

#	Short description entered	Expected category	Result
T1	Wi-Fi not working in Lab 2	Network	Pass
T2	Projector not turning on	Hardware	Pass
T3	I forgot my password	Account / Password	Pass
T4	Computer is very slow	Performance	Pass
T5	Need help with a new software request	Other (default)	Pass
T6	Keyword only in Description field	Matches by Description	Pass

Update the Result column with your actual test outcomes.

Performance checks

- Flow Executions show status **Completed** for each new incident.

- Category is populated immediately after the record is created.
- Multiple incidents created in succession are each classified correctly.

7. RESULTS

7.1 Output Screenshots

Insert your screenshots here:

- Figure 1: Flow Designer canvas showing trigger and conditions
- Figure 2: Update Record action configuration
- Figure 3: New incident form before saving
- Figure 4: Incident after creation with Category auto-filled
- Figure 5: Flow Executions page showing a completed run

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Automatic classification at creation	Keyword matching can misclassify vague descriptions
Saves time for IT agents	Cannot understand context or synonyms not listed
Consistent categorisation	Keyword lists need periodic updates
No-code, easy to maintain	Overlapping keywords (e.g. "slow internet") match the first rule only
Faster routing to the right team	Not suited to very high-volume, complex categorisation

9. CONCLUSION

The Auto Ticket Classification flow shows that a common helpdesk bottleneck can be solved with ServiceNow Flow Designer and no code. Incidents are categorised the moment they are created, which cuts manual triage, improves routing and keeps categories consistent. The solution is simple for administrators to understand and extend.

10. FUTURE SCOPE

- Automatically set **Priority** and **Assignment group** per category.
- Add subcategory classification.
- Send notifications to the assigned team or the caller.
- Move keyword lists to a table or Decision Table for easier maintenance.
- Adopt ServiceNow Predictive Intelligence for ML-based classification.
- Build a dashboard of tickets by category and resolution time.

11. APPENDIX

Source Code (if any): Not applicable; the solution is built entirely in Flow Designer (no code).

Dataset Link: Not applicable; the flow runs on live Incident table records.

GitHub & Project Demo Link: *Add your links here*